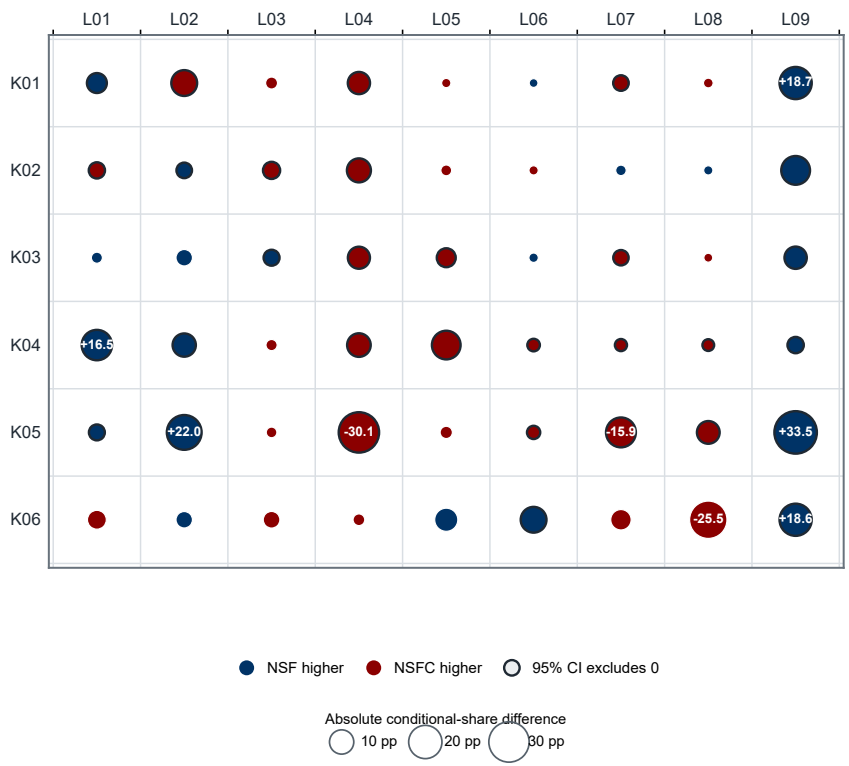
Knowledge tasks, intervention pathways and action-stage reach

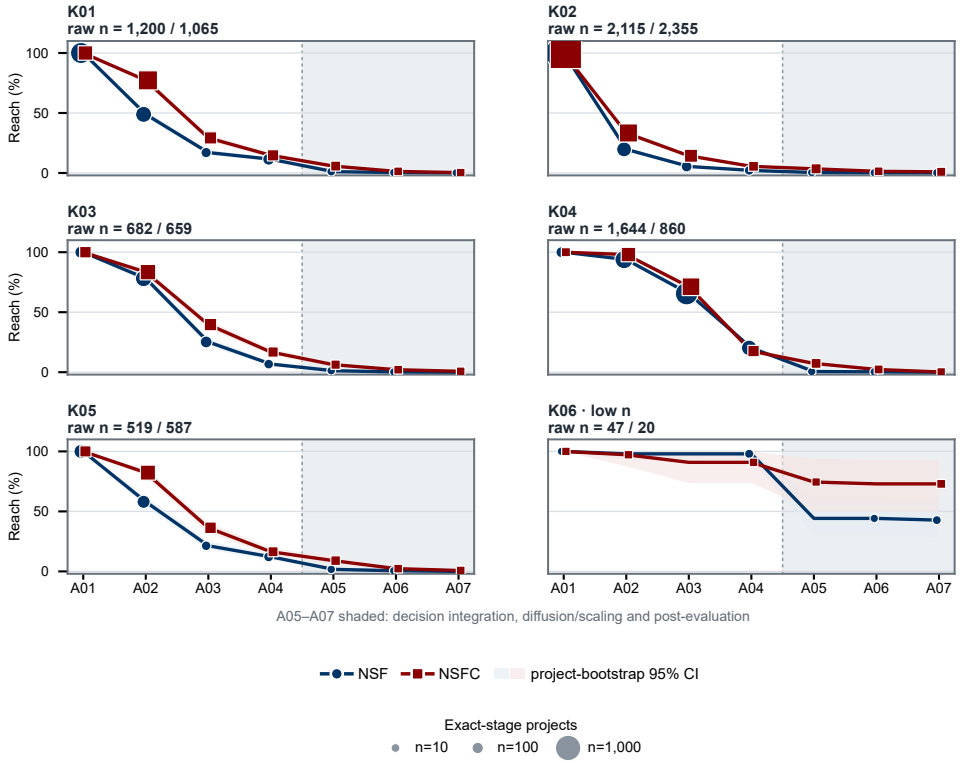
Intervention residual fingerprint

NSF−NSFC conditional share difference (percentage points)



Action-stage reach by knowledge task

curve = balanced P(stage ≥ A); bubble area = raw exact-stage project n



Knowledge tasks

K01	State measurement and problem diagnosis
K02	Mechanism analysis and causal identification
K03	Scenario simulation and risk prediction
K04	Intervention design and performance optimization
K05	Planning decision support and policy design
K06	Implementation monitoring and impact evaluation

Intervention types

L01	Engineering and infrastructure intervention	**L02**	Digital technology intervention	**L03**	Ecological restoration intervention
L04	Spatial planning intervention	**L05**	Public service and operational intervention	**L06**	Behavioural and demand-management intervention
L07	Legal and institutional intervention	**L08**	Fiscal, financial and market intervention	**L09**	Participation, collaboration and capacity-building intervention